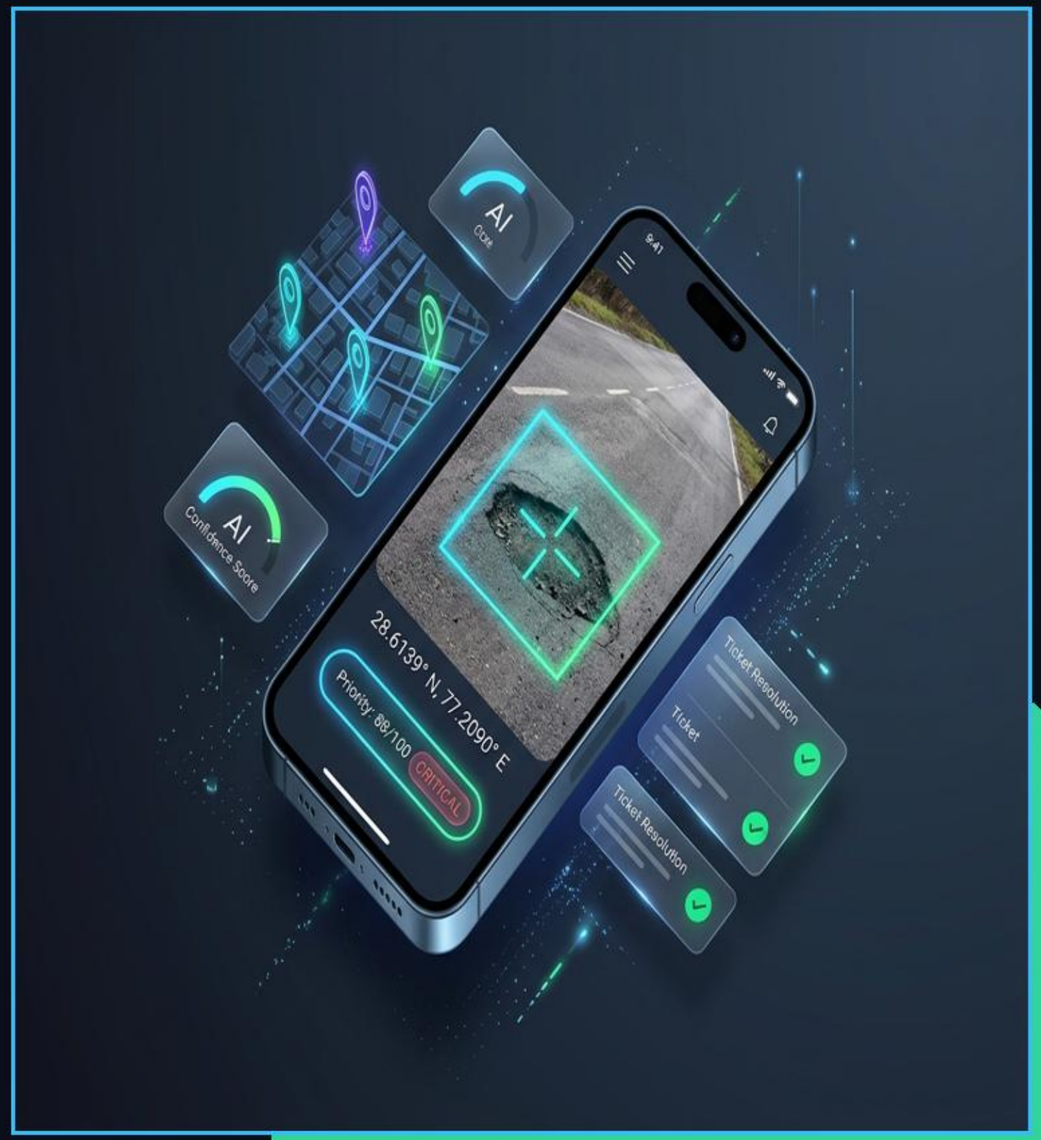
HACKNOVATE 2026 • PROBLEM STATEMENT 9

CivicPulse AI

Autonomous Civic Issue Intelligence & Transparent Resolution Protocol

Transforming civic complaints from an opaque black hole into an active, verifiable governance pipeline powered by Explainable AI Priority Scoring, Duplicate Ticket Suppression, and Before/After Proof-of-Fix.

- Explainable Dynamic Priority Scoring (0–100 EDPS Engine)
- Intelligent 'Duplicate Shield' (+1 Crowd Upvoting)
- Verifiable 'Proof-of-Fix' Protocol (CV Diff & Consensus)



• OUR PROPOSED SOLUTION

CivicPulse: The 3 Pillars of Our Solution

A balanced tri-party platform uniting Citizens, Algorithmic Prioritization, and Municipal Field Teams.

PILLAR 1: SMART REPORTING

Frictionless Edge Capture

- **Instant Geo-Tagging**
  1-click photo auto-extracts EXIF GPS coordinates & reverse geocodes the exact street address.
- **AI Multimodal Vision**
  Pre-classifies defect category (Pothole, Waste, Streetlight) with confidence score.
- **Voice-First Ingestion**
  Speak grievances naturally in Hindi or English with real-time speech transcription.
- **Zero Barriers**
  No tedious logins required. Designed for 100% citizen adoption.

PILLAR 2: ALGORITHMIC TRIAGE

Explainable Priority (EDPS)

- **Mathematical Formulation**
  Calculates a dynamic 0–100 severity index combining 5 real-time vectors.
- **Geospatial Zone Proximity**
  Distance-decay boost for defects near hospitals, schools, and transit hubs.
- **Civic Equity Guarantee**
  Prevents less digitized wards from being deprioritized due to smartphone disparity.
- **Plain-English Reasoning**
  Generates a human-readable audit card explaining why the score was assigned.

PILLAR 3: PUBLIC ACCOUNTABILITY

Verifiable Proof-of-Fix

- **Before / After AI Audit**
  Repair crews must upload an After photo at the exact GPS coordinates; CV verifies fix.
- **Community Consensus**
  Original reporter & nearby citizens vote to confirm quality before ticket archives.
- **Public Milestone Tracker**
  Full public visibility from Reported → Triaged → Dispatched → Verified.
- **Ward SLA Scorecards**
  Live ward rankings holding municipal contractors accountable.

SYSTEM WORKFLOW

How It Works: 4 Transparent Steps to Resolution

From citizen smartphone to automated triage, field repair, and public verification.

1 CAPTURE

Auto EXIF GPS
AI vision category
Voice in Hindi/Eng

2 AI TRIAGE

EDPS 0-100 score
School/hospital check
Explainability card

3 DISPATCH

Ward auto-routing
Strict SLA timer
Contractor queue

4 PROOF-OF-FIX

Geotagged after photo
CV difference audit
Citizen consensus

Step 1: Smart Capture

Citizen snaps photo. GPS coordinates, timestamp, and AI hazard bounding box are locked instantly.

Step 2: AI Triage (EDPS)

Calculates dynamic 0–100 priority score factoring in depth, hospital proximity, and crowd upvotes.

Step 3: Ward Dispatch

Auto-routed to the specific Ward Department (PWD/Sanitation) with a strict SLA countdown timer.

Step 4: Proof-of-Fix

Contractor uploads geotagged after-photo. AI diff checks repair, and citizens cast a verification vote.

• THE ALGORITHMIC CORE

Explainable Dynamic Priority Scoring (EDPS Engine)

Directly tackling the hackathon challenge: genuine mathematical prioritization with transparent reasoning.



EXPLAINABLE AI BREAKDOWN

Transparent audit trail for citizens and municipal engineers

Visual Hazard Depth (Sv)
18cm deep crater, exposed sub-base rock

Zone Proximity (Zw)
40 meters from St. Jude Hospital Emergency Bay

Cluster Velocity (Cv)
9 citizen reports in 90 mins (Surge multiplier)

Civic Equity Index (Eq)
Under-reported ward baseline parity applied

THE FORMULA: Priority = min(100, 0.30·Sv [Hazard] + 0.25·Zw [Zone] + 0.20·Cv [Velocity] + 0.15·Eq [Equity] + 0.10·Te [SLA])

• MUNICIPAL EFFICIENCY

The 'Duplicate Shield': Eliminating Alert Fatigue

How CivicPulse solves the avalanche of redundant complaints that paralyze city engineers.

TRADITIONAL PORTAL

The Duplicate Disaster

Ticket #101 Pothole	Ticket #102 Pothole	Ticket #103 Pothole	Ticket #104 Pothole
Ticket #105 Pothole	Ticket #106 Pothole	Ticket #107 Pothole	Ticket #108 Pothole
Ticket #109 Pothole	Ticket #110 Pothole	Ticket #111 Pothole	Ticket #112 Pothole

Result: 40 identical tickets jam the database

- Municipal officers waste hours manually sorting
- Leads to alert fatigue and ignored complaints

CIVICPULSE INNOVATION

The 'Duplicate Shield'

1 MASTER TICKET CREATED

GPS: Sector 14 Junction • Initial Report: 10:15 AM

THIS')

Result: 75% Less Database Bloat + Surge Alert

- Proactively prompts users within 50 meters to upvote
- Increases Cluster Velocity score while keeping data clean

IMPACT: 75% Reduction in Database Bloat • Zero Alert Fatigue • Instant Emergency Surge Detection

• PUBLIC ACCOUNTABILITY

The 'Proof-of-Fix' Protocol: Verifiable Closures

Restoring citizen trust through Computer Vision Before/After verification and community consensus.

1. REPORTED ISSUE

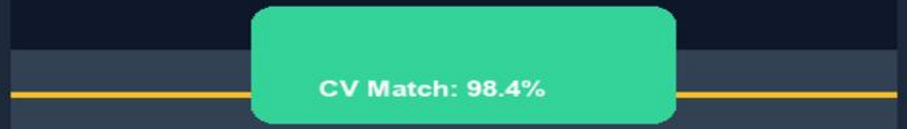


GPS: 28.6139° N, 77.2090° E (EXIF Locked)

Status: Unpaved Crater (18cm depth) • Oct 28

Assigned Priority: 88/100 (Critical)

2. RESOLVED & AUDITED



GPS Verified: Within 4.2m of coordinates

Repaired with Hot-Mix Asphalt • Contractor #PWD-41

Community Consensus: 4/4 Citizens Confirmed Fixed

CLOSING THE LOOP: Geotagged Photo Lock • AI Computer Vision Difference Analysis • 24h Community Consensus Gate

• TWO-SIDED PLATFORM

Public Resolution Tracker & Municipal Command Center

Empowering both citizens and city administration through real-time geospatial intelligence.

FOR CITIZENS: PUBLIC RESOLUTION TRACKER

100% Transparent Civic Governance

- **Live 5-Stage Journey**
  Real-time tracking: Submitted → AI Triaged → Dispatched → Repaired → Verified.
- **Interactive GIS City Map**
  Search any neighborhood or ward to see active, in-progress, and resolved repairs.
- **Explainability Cards**
  Every ticket displays the exact reason for its assigned priority score.
- **Community Voice**
  Vote on the quality of completed repairs directly from your phone.

FOR CITY OFFICIALS: COMMAND CENTER

Data-Driven Municipal Operations

- **Geospatial Hotspot Heatmap**
  Pinpoints civic failure density across urban sectors in real time.
- **Automated SLA Countdown**
  Critical tickets trigger strict 24-hour clocks with auto-escalation to commissioners.
- **Work-Order Routing**
  Automatically dispatches tickets to specific ward departments (PWD, Electricity, Sanitation).
- **Ward Performance Leaderboard**
  Ranks contractors and ward officers by fix speed, quality, and citizen satisfaction.

HACKNOVATE 2026 • PROBLEM STATEMENT 9

Thank You!

CivicPulse AI: Turning Civic Frustration into Prioritized, Transparent Civic Action.

We are open for Questions & Feedback

- AI Explainable Dynamic Priority (EDPS)
- Duplicate Shield (+1 Upvoting)
- Verifiable Proof-of-Fix Protocol

Designed & Engineered for Problem Statement 9 | Hacknovate 2026